JATIN SHARMA

CONTROL SYSTEMS ENGINEER | EMBEDDED SYSTEMS, RADIO & ROBOTICS

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Open to relocation (including Switzerland), international travel and cross-time-zone collaboration

PROFESSIONAL PROFILE

Control systems and electrical engineer with 5+ years of experience in SKAO radio telescope environments, embedded firmware, feedback control and complex systems integration. Hands-on contributions to **five-point scan-based pulsar detection/tracking**, antenna timing/delay modelling, radio-system monitoring and real-time diagnostics. Familiar with linear/nonlinear system dynamics, Kalman/EKF state-estimation concepts, dynamic programming and recursive methods; experienced in hardware-software prototyping and validation.

CORE TECHNICAL COMPETENCIES

Control systems & estimation: Applied mathematics and system dynamics; linear/nonlinear control theory; PI/PD control; Kalman and Extended Kalman filter concepts; stochastic estimation; recursive techniques; dynamic programming.

Modelling & validation: System identification and model validation concepts; feedback control design; timing/delay modelling; MATLAB/Simulink familiarity for control-system modelling and simulation.

Radio systems & sensing: Five-point pulsar scanning/tracking; antenna positioning; SKA-Low inter-antenna delays; satellite synchronization; noise-aware signal handling; dual-antenna radio-signal capture.

Embedded systems & robotics: Embedded firmware controllers; electronics prototyping; electromechanical integration; humanoid robotics; robotic arm/antenna motion; PI/PD controller implementation.

Software, integration & testing: Python, Linux, Bash, Git, GitLab CI/CD, Pytest, BDD/Gherkin, Docker, Kubernetes, Helm; Tango, EDA, HDB++; Jama requirements traceability.

PROFESSIONAL EXPERIENCE

System Engineer | Tata Consultancy Services

SKAO client engagement | Jul 2021 - Present | Pune / Remote

- **Implemented five-point scan control logic for pulsar detection and tracking** and developed embedded firmware controllers for telescope observing and monitoring workflows.
- Contributed to **SKA-Low CSP inter-antenna delay calculation code** with fitted estimates and weather-aware correction inputs for timing-sensitive radio telescope operations.
- Developed Python/Linux **monitoring and control software**, operator interfaces and real-time telemetry/diagnostics using Tango, EDA and HDB++.
- Supported cross-subsystem integration and requirements coordination; delivered Dish LMC dashboard features in an Integration Test Facility and investigated integration faults.
- Refactored legacy code for an approximately **30% performance improvement**; expanded automated validation with an approximately **80% increase in test coverage**.
- Supported integration and deployment using Pytest, BDD, Docker, Kubernetes, Helm and GitLab CI/CD; conducted on-site troubleshooting at **CSIRO, Australia (Jul 2025)**.

SELECTED CONTROL, RADIO & ROBOTICS PROJECTS

Five-Point Pulsar Scanning & Telescope Control Firmware | SKAO

- Implemented feedback-oriented controller logic for **five-point scanning used in pulsar detection/tracking**, together with embedded firmware for SKAO telescope systems.
- Integrated scanning/control functions with monitoring, subsystem diagnostics and synchronization-sensitive operating environments.

PI/PD Controller & Embedded Control Development

- Developed PI/PD control logic and embedded firmware for electromechanical systems, applying feedback control principles and practical hardware-software integration.

Robotic Arm / Stowable Radio Antenna / Signal Capture

- Designed and demonstrated an electronics-integrated robotic arm and antenna concept using **dual-antenna radio-signal capture** and a virtual-camera approach.
- Integrated motion planning, noise-aware signal handling and electronics prototyping; showcased the work at an Innovation Sprint.

GYANI 2.0 Humanoid Robot | JECRC

Mar 2018 - Apr 2019

- Developed embedded control logic, including **PI (proportional-integral) control**, and contributed to mechanical/electrical hardware-software integration for humanoid robotics.

ADDITIONAL SATELLITE & TIMING EXPERIENCE

Satellite Synchronisation Engineer | Additional Domain Exposure

Jul 2021 - Jun 2022

- Supported timing-sensitive satellite synchronisation and operational troubleshooting to preserve alignment for reliable Earth-data acquisition.

EDUCATION & RECOGNITION

B.Tech. in Electrical Engineering | Jaipur Engineering College and Research Centre, Rajasthan Technical University | 2017 - 2021

Publication | AR Platform for Indoor Navigation: New Potential Approach Extensible to Older People with Cognitive Impairment

Recognition | Six consecutive Innovation Prize / Code Quality Prize recognitions in the SKAO environment (from Dec 2024); Best Out-of-the-Box Project, Innovation Sprint (Mar 2026).

ENGINEERING PORTFOLIO & DEMONSTRATIONS

GitHub: github.com/rootjatin | SKAO engineering GitLab: gitlab.com/jatin.ska

SKA CSP contribution: SKA-Low CSP leaf-node MR 154 (timing-delay calculations)

Humanoid robot demo: youtube.com/watch?v=bUHOEZJiqfk

Robotic antenna / pulsar demo: youtube.com/watch?v=b2XdSoMZLQc